PROJECT REVIEW 1

Multimodal Hallucination Firewall

Evidence-Based Hallucination Detection for Multimodal RAG Systems

Project Theme	Combining multimodal AI, retrieval-augmented generation, and machine learning to detect unsupported or contradictory AI-generated responses using visual and textual evidence.
Course	M.Sc. Artificial Intelligence & Machine Learning
Subject	Machine Learning
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1. Abstract

Generative AI systems can produce fluent and confident responses containing information that is not supported by available evidence. This challenge becomes more difficult in multimodal systems, where information may be distributed across images, text, and retrieved documents.

This project proposes a **Multimodal Hallucination Firewall**, an AI-assisted verification framework for Retrieval-Augmented Generation (RAG) systems. The framework will accept multimodal inputs such as an image and a text query, retrieve relevant supporting information, generate an answer, and then verify whether important claims are supported by visual and textual evidence.

The central research direction is to combine semantic similarity, image-text alignment, claim verification, and machine-learning-based risk classification to distinguish grounded responses from potentially hallucinated responses.

Core Research Idea: Can multimodal evidence and machine-learning-based verification improve detection of unsupported claims in RAG-generated answers?

2. Keywords

Keywords: Multimodal AI; Hallucination Detection; RAG; Generative AI; Computer Vision; Natural Language Processing; Vision-Language Models; Evidence Verification; Machine Learning; Responsible AI.

3. Data Description

The project will use multimodal samples containing an image, a text question or prompt, supporting evidence where applicable, and an answer or claim to be verified. The final dataset/source will be selected based on suitability for multimodal hallucination and faithfulness evaluation.

3.1. Basic Data Preparation Process

Image + Question → Evidence Retrieval → Multimodal Answer Generation → Claim Extraction → Evidence Verification → Grounded / Hallucinated Label

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3.2. Dataset Characteristics

Parameter	Description
Data Type	Multimodal image-text and evidence-based data
Primary Modalities	Image + text; retrieved textual evidence may also be used
Input	Image and user question / prompt
Evidence	Retrieved or associated textual and visual information
Learning Target	Grounded / unsupported / contradictory response or hallucination-risk score
Key Features	Text similarity, image-text alignment, entailment, entity and numerical consistency

4. Proposed Verification Framework

The firewall operates after answer generation and evaluates the response against the evidence available to the system. Multiple complementary signals will be investigated instead of relying on a single similarity score.

Verification Signal	Purpose
Semantic Similarity	Measures agreement between the answer and retrieved textual evidence.
Image-Text Alignment	Measures compatibility between visual content and textual claims.
Claim Verification / NLI	Checks whether evidence supports, contradicts, or does not support a claim.
Entity Consistency	Checks important objects, names, locations and attributes.
Numerical Consistency	Checks quantities, dates, prices, percentages and specifications.

Illustrative example: If a product image and supporting specification identify a processor as Intel Core i7, but the generated answer states Intel Core i9, the firewall should identify the contradiction and assign a higher hallucination risk.

5. Outcomes

The project is intended to produce the following outcomes:

1. Develop a multimodal RAG environment for evidence-grounded question answering.
2. Build a verification layer for detecting unsupported or contradictory generated claims.
3. Investigate whether combining visual and textual evidence improves hallucination detection.
4. Develop machine-learning features and a classifier/risk-scoring mechanism for response verification.
5. Compare verification signals and study their contribution to detection performance.

6. Develop an interactive prototype demonstrating answer generation, evidence checking and hallucination-risk scoring.

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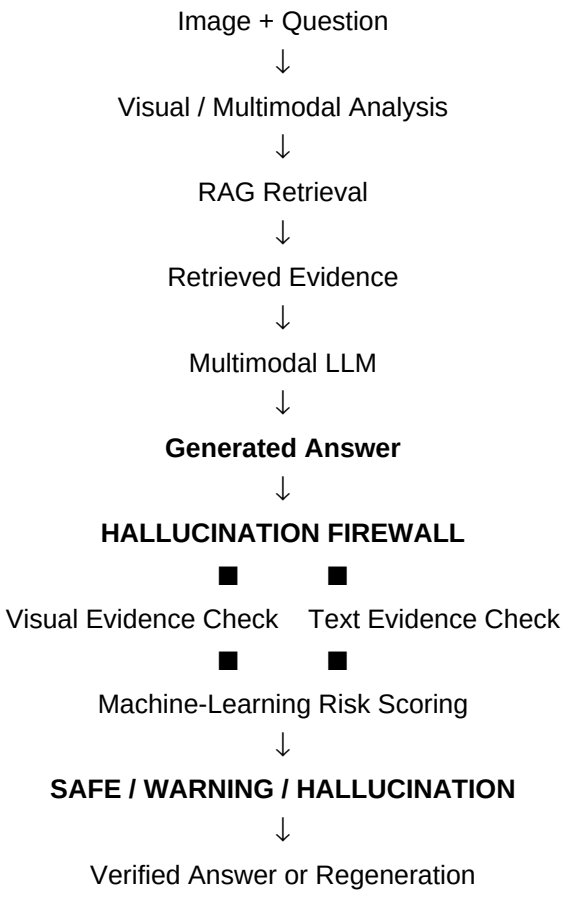
6. Project Direction

The overall direction of the proposed system can be summarized as:



7. Project Focus

The project will investigate whether multimodal evidence verification can improve the reliability of RAG-generated answers. Particular attention will be given to the relationship between visual evidence, retrieved textual evidence, generated claims, and the final hallucination-risk decision.

8. Expected Project Contribution

The expected contribution is a hybrid framework combining **multimodal evidence, RAG, and machine learning** to identify potentially hallucinated responses. The system is intended to provide not only an answer, but also an interpretable verification outcome indicating why a response is considered grounded or risky.

9. Initial Technology Direction

Component	Possible Direction
Vision / Vision-Language	CLIP, BLIP/BLIP-2, or a suitable vision-language model
Text Representation	Sentence-transformer or comparable embeddings
Retrieval	FAISS or another vector-search mechanism
ML Classifier	Logistic Regression, Random Forest, SVM or XGBoost

Evaluation	Accuracy, Precision, Recall, F1-score and ROC-AUC

Project Objective: To develop and evaluate a multimodal AI verification layer that reduces the risk of unsupported information being presented as a reliable answer.

Note: Detailed model architecture, final dataset selection, experimental comparisons, quantitative results, and the complete prototype will be developed and presented in subsequent project reviews.